CALCULUS WITH

Asante

TOPIC TWO | TOPIC SERIES

Limits

Compiled and taught by Asante

MATH 152 | Calculus & Analysis | KNUST BME1

Why This Booklet Exists

This booklet is here to supplement your own learning — not replace it. Lectures, your notes, your textbook, and the time you spend wrestling with a problem on your own all still matter. Nothing here is a shortcut around doing the work; it's a second angle on the same material, laid out the way I wish someone had laid it out for me.

Learning is not a straight line. You will understand the Squeeze Theorem on page 14, forget it exists two weeks later, and only really get it when it shows up again in a totally different problem. That's normal. This booklet is built to be revisited, not read once and shelved.

If you get stuck, that's the point where the real learning starts, chale — not a sign that something is wrong with you. Sit with the confusion for a bit before you flip to the answer.

Every booklet in this series also closes with a Cheat Sheet — a fast-reference summary of the whole topic. It's there for review the night before a test, not for learning something for the first time.

A Note on How to Use This

Don't try to read this cover to cover in one sitting the night before a test. Work through one section, attempt the boxed questions honestly, and come back to whatever didn't stick.

How This Booklet Works

Every topic in this series is laid out the same way, so once you learn the pattern once, you can move faster through every booklet after this one. Here's what each marker means.

TRY THIS FIRST

A few questions up front, before any explanation. Attempt them cold — they tell you exactly which part of this page you actually need.

DEFINITION

The formal statement — rule, theorem, or term — stated precisely. Worth memorising word for word.

EXAMPLE

A worked problem, solved in full, showing the reasoning step by step, not just the answer.

TRY IT

Your turn. Attempt it before checking the answer given underneath.

RECALL / STRATEGY

A shortcut, common trap, or the exact test to reach for — the thing you should be thinking right before you attempt a problem.

WATCH & LEARN

Hyperlinked videos, dropped in right next to the concept they explain — click (▶) whenever reading alone isn't landing.

Each booklet ends with a Problem Set and a Cheat Sheet summary. Questions marked **T** are technology-appropriate — a calculator or CAS is fine for checking arithmetic, not for skipping the reasoning. Full worked solutions live in the separate Answer Key that accompanies this booklet.

Asante Inc. understands that learning is not a linear process and can be stressful — so this booklet has hyperlinked videos built in throughout, plus a short list of videos worth watching before you even start reading. Look for ▶ next to any concept that isn't landing.

Key Vocabulary

A quick glossary before you dive in. Come back here any time a term stops making sense mid-page.

Limit

The value a function approaches as its input approaches some point — like watching your speedometer as you roll up to a red light. You might stop just before hitting exactly 0 km/h, but the reading is clearly heading there.

One-Sided Limit

The value approached from only one direction — left or right. Like judging a temperature trend using only the morning readings, ignoring the afternoon.

Limit at Infinity

What a function settles toward as x grows without bound. Think of a savings balance under a capped interest scheme — it keeps growing but never clears the ceiling.

Horizontal Asymptote

The flat line a graph hugs as $x \to \pm\infty$ — the visual signature of a limit at infinity.

Indeterminate Form

An expression like $\frac{0}{0}$ or $\frac{\infty}{\infty}$ that doesn't yet tell you the answer — like being told "the match ended level" without being told the actual score.

Key Vocabulary (continued)

Continuity

A function with no breaks, jumps, or holes over an interval — a road with no potholes or missing sections.

Discontinuity

A break in the graph: removable (a single missing point, easily patched), jump (an abrupt step, like a taxi fare that jumps the instant you cross a mile marker), or infinite (the function blows up, like $1/x$ at $x = 0$).

Squeeze Theorem

If a function is trapped between two others that converge to the same limit, it's forced to that same limit too — like being sandwiched between two friends walking at matching pace toward the same door.

Intermediate Value Theorem

If a continuous function takes two values, it must take every value in between somewhere — a car accelerating from 0 to 100 km/h has to pass through 50 km/h at some exact instant.

Recommended Videos --- Watch Before Reading

You don't need to watch all the videos in this booklet before starting — just these five. They cover the backbone ideas that everything else in this topic builds on.

▶ Intro to Limits

▶ Evaluating Limits

▶ How to Find Limits at Infinity

▶ Squeeze Theorem

▶ Continuity and Differentiability

Every video used across this booklet — and every video still to come for future topics — lives in one running playlist:

▶ Full Playlist — Limits, Calculus with Asante

Try This First --- 5 Questions, No Notes

1. Find $\lim_{x \to 2} (x^2 + 1)$.
   (A) 3
   (B) 5
   (C) 4
   (D) 6

2. Find $\lim_{x \to 0} \frac{\sin x}{x}$.
   (A) 0
   (B) Undefined
   (C) 1
   (D) ∞

3. Does $\lim_{x \to 0} \frac{|x|}{x}$ exist?
   (A) Yes, it equals 1
   (B) Yes, it equals -1
   (C) Yes, it equals 0
   (D) No — the one-sided limits disagree

4. Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.
   (A) $\frac{3}{5}$
   (B) $\frac{2}{5}$
   (C) 1
   (D) ∞

5. Is $f(x) = |x|$ differentiable at $x = 0$?
   (A) Yes, the derivative is 0
   (B) Yes, the derivative is 1
   (C) No — the left- and right-hand slopes disagree
   (D) No — f is discontinuous there

HOW DID THAT FEEL?

Questions 3 and 5 are the ones that catch people — both hinge on the same idea: a function can be perfectly well-behaved and still have a sharp corner or a split personality at one exact point. That idea runs through this entire booklet.

1. Intro to Limits

DEFINITION --- LIMIT

$\lim_{x \to a} f(x) = L$ means: as x gets arbitrarily close to a (from either side), $f(x)$ gets arbitrarily close to L. Crucially, $f(a)$ itself doesn't have to exist or equal L — the limit only cares about the approach.

Example 1

Find $\lim_{x \to 2}(x^2 + 1)$.

Solution. $f(x) = x^2 + 1$ is continuous everywhere, so direct substitution works: $f(2) = 4 + 1 = 5$.

▶ Watch: Intro to Limits

2. Limits of Radical Functions

RECALL / STRATEGY --- RATIONALIZE FIRST

Direct substitution into a radical limit often gives $\frac{0}{0}$. Multiply by the conjugate to clear the root before substituting again.

Example 2

Find $\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$.

Solution. Direct substitution gives $\frac{0}{0}$. Multiply by the conjugate $\sqrt{x} + 2$:

$$\frac{\sqrt{x} - 2}{x - 4} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2} = \frac{x - 4}{(x - 4)(\sqrt{x} + 2)} = \frac{1}{\sqrt{x} + 2}$$

Now substitute: $\lim_{x \to 4} \dfrac{1}{\sqrt{x} + 2} = \dfrac{1}{4}$.

▶ Watch: Evaluating Limits

3. Limits at Infinity

DEFINITION --- LIMIT AT INFINITY

$\lim_{x \to \infty} f(x) = L$ means $f(x)$ settles toward L as x grows without bound. If L is finite, $y = L$ is a horizontal asymptote.

Example 3

Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.

Solution. Divide top and bottom by x^2, the highest power present:

$$\lim_{x \to \infty} \frac{3 + \frac{2}{x}}{5 - \frac{1}{x^2}} = \frac{3}{5}$$

The graph has a horizontal asymptote at $y = \frac{3}{5}$.

▶ Watch: How to Find Limits at Infinity

4. Limits and Trig Functions

RECALL --- THE TWO FOUNDATIONAL TRIG LIMITS

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \qquad \lim_{x \to 0} \frac{1 - \cos x}{x} = 0$$

These two limits are the building blocks for every trig-derivative proof later on — worth memorising outright.

Example 4

Confirm $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$ makes sense numerically: as $x \to 0$, $\cos x \to 1$, so the numerator shrinks to 0 faster than x does, and the ratio flattens to 0.

▶ Watch: Limits of Trig Functions

▶ Watch: More Trig Limit Practice

5. Limits and Absolute Value

RECALL / STRATEGY --- SPLIT IT BY SIGN

$|x|$ behaves differently depending on the sign of x: $|x| = x$ for $x \geq 0$, and $|x| = -x$ for $x < 0$. Always check both one-sided limits separately.

Example 5

Does $\lim_{x \to 0} \frac{|x|}{x}$ exist?

Solution.

$$\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} = 1 \qquad \lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} = -1$$

The one-sided limits disagree ($1 \neq -1$), so the two-sided limit does not exist.

▶ Watch: Limits and Absolute Value

6. Limits of Logarithmic Functions

DEFINITION --- BEHAVIOUR OF LOGS NEAR 0 AND BEYOND

For $\log_b x$ with $b > 1$: as $x \to 0^+$, $\log_b x \to -\infty$ (a vertical asymptote at $x = 0$); as $x \to \infty$, $\log_b x \to \infty$, but slower than any positive power of x.

Example 6

Find $\lim_{x \to 100} \log_{10} x$.

Solution. $\log_{10} x$ is continuous for $x > 0$, so direct substitution works: $\log_{10} 100 = 2$.

The natural log $\ln x$ is the special case $b = e$ — important enough to get its own section next.

7. Limits and Natural Logarithms

Example 7

Find $\lim_{h \to 0} \dfrac{\ln(1+h)}{h}$.

Solution. Direct substitution gives $\frac{0}{0}$ — but this exact limit is the definition of the derivative of $\ln x$ at $x = 1$, and it evaluates to

$$\lim_{h \to 0} \frac{\ln(1+h)}{h} = 1$$

This one resurfaces constantly once derivatives are on the table — worth remembering by name.

▶ Watch: Limits and Natural Logarithms

8. Squeeze Theorem

DEFINITION --- SQUEEZE (SANDWICH) THEOREM

If $g(x) \leq f(x) \leq h(x)$ near a, and $\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$, then $\lim_{x \to a} f(x) = L$ too — f has no choice but to follow.

Example 8

Find $\lim_{x \to 0} x^2 \sin\left(\dfrac{1}{x}\right)$.

Solution. $\sin(1/x)$ oscillates wildly near 0, but it's always trapped: $-1 \leq \sin(1/x) \leq 1$. Multiply through by $x^2 \geq 0$:

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

Both outer bounds $\to 0$ as $x \to 0$, so by the Squeeze Theorem, $\lim_{x \to 0} x^2 \sin\left(\dfrac{1}{x}\right) = 0$.

▶ Watch: Squeeze Theorem

9. Limits of Piecewise Functions

RECALL / STRATEGY --- CHECK BOTH SIDES AT THE SEAM

For a piecewise function, the limit at a breakpoint only exists if the left-hand and right-hand limits agree. Always evaluate each piece separately using the formula valid on that side.

Example 9

Let $f(x) = \begin{cases} x+1 & x<2 \\ 3 & x=2 \\ x^2-1 & x>2 \end{cases}$. Find $\lim_{x\to 2} f(x)$.

Solution.

$$\lim_{x\to 2^-} f(x) = 2+1 = 3 \qquad \lim_{x\to 2^+} f(x) = 2^2-1 = 3$$

Both sides agree, so $\lim_{x\to 2} f(x) = 3$. Since this also equals $f(2)=3$, f is continuous at $x=2$.

TRY IT --- YOUR TURN

Let $g(x) = \begin{cases} x^2 & x<1 \\ 3 & x=1 \\ 2x+1 & x>1 \end{cases}$. Does $\lim_{x\to 1} g(x)$ exist?

Answer: No — $\lim_{x\to 1^-} g(x) = 1$ but $\lim_{x\to 1^+} g(x) = 3$. The one-sided limits disagree, so g has a jump discontinuity at $x=1$, regardless of what $g(1)$ is defined to be.

▶ Watch: Limits of Piecewise Functions

▶ Watch: More Piecewise Practice

10. Limits of Inverse Trig Functions

DEFINITION --- DOMAIN LIMITS

Inverse trig functions are only defined on restricted domains. $\cos^{-1} x$ and $\sin^{-1} x$ only exist for $x \in [-1, 1]$, so limits approaching the domain's edge are automatically one-sided.

Example 10

Find $\lim_{x \to 1^-} \cos^{-1} x$.

Solution. $\cos^{-1} x$ is continuous on $[-1, 1]$, so direct substitution from within the domain works: $\lim_{x \to 1^-} \cos^{-1} x = \cos^{-1}(1) = 0$.

TRY IT --- YOUR TURN

Find $\lim_{x \to \infty} \tan^{-1} x$ and $\lim_{x \to -\infty} \tan^{-1} x$.

Answer: $\frac{\pi}{2}$ and $-\frac{\pi}{2}$ — unlike $\cos^{-1}$ and $\sin^{-1}$, $\tan^{-1}$ is defined for all real x and has two horizontal asymptotes, one on each side.

▶ Watch: Limits of Inverse Trig Functions

11. Intermediate Value Theorem

DEFINITION --- INTERMEDIATE VALUE THEOREM (IVT)

If f is continuous on $[a, b]$ and k is any value between $f(a)$ and $f(b)$, then there exists at least one $c \in (a, b)$ with $f(c) = k$. Continuity is what guarantees the function can't "jump over" k.

Example 11

Show $f(x) = x^3 - x - 1$ has a root between $x = 1$ and $x = 2$.

Solution. f is a polynomial, so it's continuous everywhere. Check the endpoints:

$$f(1) = 1 - 1 - 1 = -1 \qquad f(2) = 8 - 2 - 1 = 5$$

Since $f(1) < 0 < f(2)$, by the IVT there exists some $c \in (1, 2)$ with $f(c) = 0$ — a root, even though we haven't found its exact value.

12. Continuity & Differentiability --- Intro

DEFINITION --- CONTINUITY AT A POINT

f is continuous at $x = a$ if all three hold: (1) $f(a)$ exists, (2) $\lim_{x \to a} f(x)$ exists, and (3) they're equal. Miss any one of the three, and f is discontinuous at a.

RECALL / STRATEGY --- DIFFERENTIABLE $\Rightarrow$ CONTINUOUS, NOT THE REVERSE

Every differentiable function is continuous. But continuity does not guarantee differentiability — a function can have no breaks and still have a sharp corner.

Example 12

Show $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there.

Solution. Continuity: $f(0) = 0$, and $\lim_{x \to 0} |x| = 0$ from both sides — so f is continuous at 0. *Differentiability:* the slope from the right is $\lim_{h \to 0^+} \frac{|h|}{h} = 1$; from the left it's $\lim_{h \to 0^-} \frac{|h|}{h} = -1$. The two sides disagree, so no single tangent slope exists at $x = 0$ — f is not differentiable there, despite being perfectly continuous.

This is exactly as deep as continuity and differentiability need to go for now — they've earned a full booklet of their own, coming soon in this series.

▶ Watch: Continuity and Differentiability

▶ Watch: More on Continuity & Differentiability

▶ Watch: Test for Continuity

Putting It Together

- A limit describes what $f(x)$ approaches, not necessarily what it equals at that point.
- Radical limits: rationalize with the conjugate before substituting.
- Limits at infinity: divide by the highest power to find horizontal asymptotes.
- $\lim_{x \to 0} \frac{\sin x}{x} = 1$ and $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$ are foundational — memorise both.
- Absolute value and piecewise limits: always check left and right separately; they must agree for the limit to exist.
- Logs: $\log_b x \to -\infty$ as $x \to 0^+$; $\lim_{h \to 0} \frac{\ln(1+h)}{h} = 1$ is worth knowing by name.
- Squeeze Theorem: trap the function between two others with the same limit.
- IVT: continuity guarantees a function hits every value between two known outputs.
- Differentiable $\Rightarrow$ continuous, but not the reverse — $|x|$ at 0 is the standard counterexample.

Problem Set

Questions marked **T** are technology-appropriate — a calculator or CAS is fine for checking your arithmetic, not for skipping the reasoning.

1. Find $\lim_{x \to 9} \frac{\sqrt{x} - 3}{x - 9}$.
   (A) $\frac{1}{3}$
   (B) $\frac{1}{6}$
   (C) 6
   (D) 0

2. Find $\lim_{x \to \infty} \frac{4x^3 - x}{2x^3 + 7}$.
   (A) 2
   (B) 4
   (C) $\frac{1}{2}$
   (D) ∞

3. Find $\lim_{x \to 0} \frac{\sin(3x)}{x}$.
   (A) 1
   (B) $3x$
   (C) 3
   (D) $\frac{1}{3}$

4. Does $\lim_{x \to 0} \frac{|x|}{x^2}$ exist? Explain.
   (A) Yes, it equals 0
   (B) Yes, it equals 1
   (C) No — it diverges to $+\infty$ from both sides
   (D) No — it oscillates

5. **T** Find $\lim_{x \to 1} \ln x$.
   (A) 1

(B) 0
(C) Undefined
(D) e

6. Use the Squeeze Theorem to find $\lim_{x\to 0} x^4 \cos\left(\frac{1}{x}\right)$.

   (A) 0
   (B) 1
   (C) Does not exist
   (D) ∞

7. For $h(x) = \begin{cases} 2x & x \le 3 \\ x+4 & x > 3 \end{cases}$, does $\lim_{x\to 3} h(x)$ exist?

   (A) Yes, it equals 6
   (B) Yes, it equals 7
   (C) No — the one-sided limits disagree
   (D) Yes, it equals 6.5

8. Find $\lim_{x\to -1^+} \sin^{-1} x$.

   (A) $\frac{\pi}{2}$
   (B) $-\frac{\pi}{2}$
   (C) 0
   (D) Undefined

9. Show $f(x) = \cos x - x$ has a root between $x = 0$ and $x = 1$.

   (A) Yes, by the IVT, since $f(0) > 0 > f(1)$
   (B) No — f is not continuous
   (C) Yes, but only because $f(0) = f(1)$
   (D) Cannot be determined

10. **T** Is $f(x) = \sqrt[3]{x}$ continuous at $x = 0$? Is it differentiable there?

   (A) Continuous and differentiable
   (B) Continuous, but not differentiable — vertical tangent
   (C) Not continuous, and not differentiable
   (D) Not continuous, but differentiable

Cheat Sheet

The pages that follow are Paul Dawkins's widely used Calculus Cheat Sheet, included here as a fast-reference summary — not a replacement for working through this booklet. All credit for this reference sheet belongs to its original author.

Limits

Definitions

Precise Definition : We say $\lim\limits_{x\to a} f(x) = L$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that whenever $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.

"Working" Definition : We say $\lim\limits_{x\to a} f(x) = L$ if we can make $f(x)$ as close to L as we want by taking x sufficiently close to a (on either side of a) without letting $x = a$.

Right hand limit : $\lim\limits_{x\to a^+} f(x) = L$. This has the same definition as the limit except it requires $x > a$.

Left hand limit : $\lim\limits_{x\to a^-} f(x) = L$. This has the same definition as the limit except it requires $x < a$.

Limit at Infinity : We say $\lim\limits_{x\to\infty} f(x) = L$ if we can make $f(x)$ as close to L as we want by taking x large enough and positive.

There is a similar definition for $\lim\limits_{x\to-\infty} f(x) = L$ except we require x large and negative.

Infinite Limit : We say $\lim\limits_{x\to a} f(x) = \infty$ if we can make $f(x)$ arbitrarily large (and positive) by taking x sufficiently close to a (on either side of a) without letting $x = a$.

There is a similar definition for $\lim\limits_{x\to a} f(x) = -\infty$ except we make $f(x)$ arbitrarily large and negative.

Relationship between the limit and one-sided limits

$$\lim_{x\to a} f(x) = L \quad\Rightarrow\quad \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L \qquad\qquad \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L \quad\Rightarrow\quad \lim_{x\to a} f(x) = L$$

$$\lim_{x\to a^+} f(x) \neq \lim_{x\to a^-} f(x) \quad\Rightarrow\quad \lim_{x\to a} f(x) \text{ Does Not Exist}$$

Properties

Assume $\lim\limits_{x\to a} f(x)$ and $\lim\limits_{x\to a} g(x)$ both exist and c is any number then,

1. $\lim\limits_{x\to a}[cf(x)] = c\lim\limits_{x\to a} f(x)$
2. $\lim\limits_{x\to a}[f(x) \pm g(x)] = \lim\limits_{x\to a} f(x) \pm \lim\limits_{x\to a} g(x)$
3. $\lim\limits_{x\to a}[f(x)g(x)] = \lim\limits_{x\to a} f(x)\ \lim\limits_{x\to a} g(x)$
4. $\lim\limits_{x\to a}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)}$ provided $\lim\limits_{x\to a} g(x) \neq 0$
5. $\lim\limits_{x\to a}[f(x)]^n = \left[\lim\limits_{x\to a} f(x)\right]^n$
6. $\lim\limits_{x\to a}\left[\sqrt[n]{f(x)}\right] = \sqrt[n]{\lim\limits_{x\to a} f(x)}$

Basic Limit Evaluations at $\pm\infty$

1. $\lim\limits_{x\to\infty} \mathbf{e}^x = \infty$ & $\lim\limits_{x\to-\infty} \mathbf{e}^x = 0$
2. $\lim\limits_{x\to\infty} \ln(x) = \infty$ & $\lim\limits_{x\to 0^+} \ln(x) = -\infty$
3. If $r > 0$ then $\lim\limits_{x\to\infty} \dfrac{b}{x^r} = 0$
4. If $r > 0$ and x^r is real for negative x then $\lim\limits_{x\to-\infty} \dfrac{b}{x^r} = 0$
5. n even : $\lim\limits_{x\to\pm\infty} x^n = \infty$
6. n odd : $\lim\limits_{x\to\infty} x^n = \infty$ & $\lim\limits_{x\to-\infty} x^n = -\infty$
7. n even : $\lim\limits_{x\to\pm\infty} a\,x^n + \cdots + b\,x + c = \operatorname{sgn}(a)\infty$
8. n odd : $\lim\limits_{x\to\infty} a\,x^n + \cdots + b\,x + c = \operatorname{sgn}(a)\infty$
9. n odd : $\lim\limits_{x\to-\infty} a\,x^n + \cdots + c\,x + d = -\operatorname{sgn}(a)\infty$

Note : $\operatorname{sgn}(a) = 1$ if $a > 0$ and $\operatorname{sgn}(a) = -1$ if $a < 0$.

Evaluation Techniques

Continuous Functions

If $f(x)$is continuous at a then $\lim\limits_{x\to a} f(x) = f(a)$

Continuous Functions and Composition

$f(x)$ is continuous at b and $\lim\limits_{x\to a} g(x) = b$ then

$$\lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right) = f(b)$$

Factor and Cancel

$$\lim_{x\to 2}\frac{x^2+4x-12}{x^2-2x} = \lim_{x\to 2}\frac{(x-2)(x+6)}{x(x-2)}$$
$$= \lim_{x\to 2}\frac{x+6}{x} = \frac{8}{2} = 4$$

Rationalize Numerator/Denominator

$$\lim_{x\to 9}\frac{3-\sqrt{x}}{x^2-81} = \lim_{x\to 9}\frac{3-\sqrt{x}}{x^2-81}\,\frac{3+\sqrt{x}}{3+\sqrt{x}}$$
$$= \lim_{x\to 9}\frac{9-x}{(x^2-81)(3+\sqrt{x})} = \lim_{x\to 9}\frac{-1}{(x+9)(3+\sqrt{x})}$$
$$= \frac{-1}{(18)(6)} = -\frac{1}{108}$$

Combine Rational Expressions

$$\lim_{h\to 0}\frac{1}{h}\left(\frac{1}{x+h}-\frac{1}{x}\right) = \lim_{h\to 0}\frac{1}{h}\left(\frac{x-(x+h)}{x(x+h)}\right)$$
$$= \lim_{h\to 0}\frac{1}{h}\left(\frac{-h}{x(x+h)}\right) = \lim_{h\to 0}\frac{-1}{x(x+h)} = -\frac{1}{x^2}$$

L'Hospital's/L'Hôpital's Rule

If $\lim\limits_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{0}{0}$ or $\lim\limits_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{\pm\infty}{\pm\infty}$ then,

$\lim\limits_{x\to a}\dfrac{f(x)}{g(x)} = \lim\limits_{x\to a}\dfrac{f'(x)}{g'(x)}$, a is a number, ∞ or $-\infty$

Polynomials at Infinity

$p(x)$ and $q(x)$ are polynomials. To compute $\lim\limits_{x\to\pm\infty}\dfrac{p(x)}{q(x)}$ factor largest power of x in $q(x)$ out of both $p(x)$ and $q(x)$ then compute limit.

$$\lim_{x\to-\infty}\frac{3x^2-4}{5x-2x^2} = \lim_{x\to-\infty}\frac{x^2\left(3-\frac{4}{x^2}\right)}{x^2\left(\frac{5}{x}-2\right)}$$
$$= \lim_{x\to-\infty}\frac{3-\frac{4}{x^2}}{\frac{5}{x}-2} = -\frac{3}{2}$$

Piecewise Function

$$\lim_{x\to-2} g(x) \text{ where } g(x) = \begin{cases} x^2+5 & \text{if } x < -2 \\ 1-3x & \text{if } x \ge -2 \end{cases}$$

Compute two one sided limits,

$$\lim_{x\to-2^-} g(x) = \lim_{x\to-2^-} x^2+5 = 9$$
$$\lim_{x\to-2^+} g(x) = \lim_{x\to-2^+} 1-3x = 7$$

One sided limits are different so $\lim\limits_{x\to-2} g(x)$ doesn't exist. If the two one sided limits had been equal then $\lim\limits_{x\to-2} g(x)$ would have existed and had the same value.

Some Continuous Functions

Partial list of continuous functions and the values of x for which they are continuous.

1. Polynomials for all x.
2. Rational function, except for x's that give division by zero.
3. $\sqrt[n]{x}$ (n odd) for all x.
4. $\sqrt[n]{x}$ (n even) for all $x \ge 0$.
5. $\mathbf{e}^x$ for all x.
6. $\ln(x)$ for $x > 0$.
7. $\cos(x)$ and $\sin(x)$ for all x.
8. $\tan(x)$ and $\sec(x)$ provided $x \ne \cdots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \cdots$
9. $\cot(x)$ and $\csc(x)$ provided $x \ne \cdots, -2\pi, -\pi, 0, \pi, 2\pi, \cdots$

Intermediate Value Theorem

Suppose that $f(x)$ is continuous on $[a,b]$ and let M be any number between $f(a)$ and $f(b)$. Then there exists a number c such that $a < c < b$ and $f(c) = M$.